Cellular neighbourhood (CN) analysis

Nandhu_version_final cell types · Nolan-lab method (Schürch et al., Cell 2020) · k = 10 and 20 · 10 CNs · 49 ROIs from 11 patients (6 non-relapse, 5 relapse) · 2026-11-11

Method

• **Windows.** For each cell, the window is its k nearest cells in the same ROI, including itself. Both k = 10 (about the immediate 1–2 cell-layer neighbourhood) and k = 20 (a wider niche) were run.

• **Neighbourhoods.** The cell-type fractions in each window are clustered with MiniBatchKMeans into 10 CNs (random_state 0), following the NeighborhoodCoordination repository.

• **Cell types used.** 17 annotated types from Nandhu_version_final; tumour = NPM1_blasts. Unassigned/Other and Artifact (15,462 cells) are excluded, as you chose. CNs are numbered by size (CN0 = largest). Each CN is named after its two most enriched cell types among those making up at least 5% of it.

• **Statistics.** CN abundance is the % of a ROI's cells. Non-relapse and relapse are compared per ROI (23 vs 26, Mann–Whitney, BH) and per patient (each patient = mean of its ROIs, 6 vs 5). The patient-level test is the stricter one. With 6 vs 5 patients, the smallest possible p is 0.0043.

Key findings

• **A tumour-dominated neighbourhood marks relapse.** At k = 10, CN0 (NPM1 blasts + Granulocyte) covers 30.3% of relapse versus 7.1% of non-relapse tissue (ROI BH 0.001). At k = 20 the tumour niche splits into two CNs: CN1 (NPM1 blasts + Macrophages) and CN2 (NPM1 blasts + Granulocyte). Both are expanded in relapse (ROI BH 0.004 and 0.002).

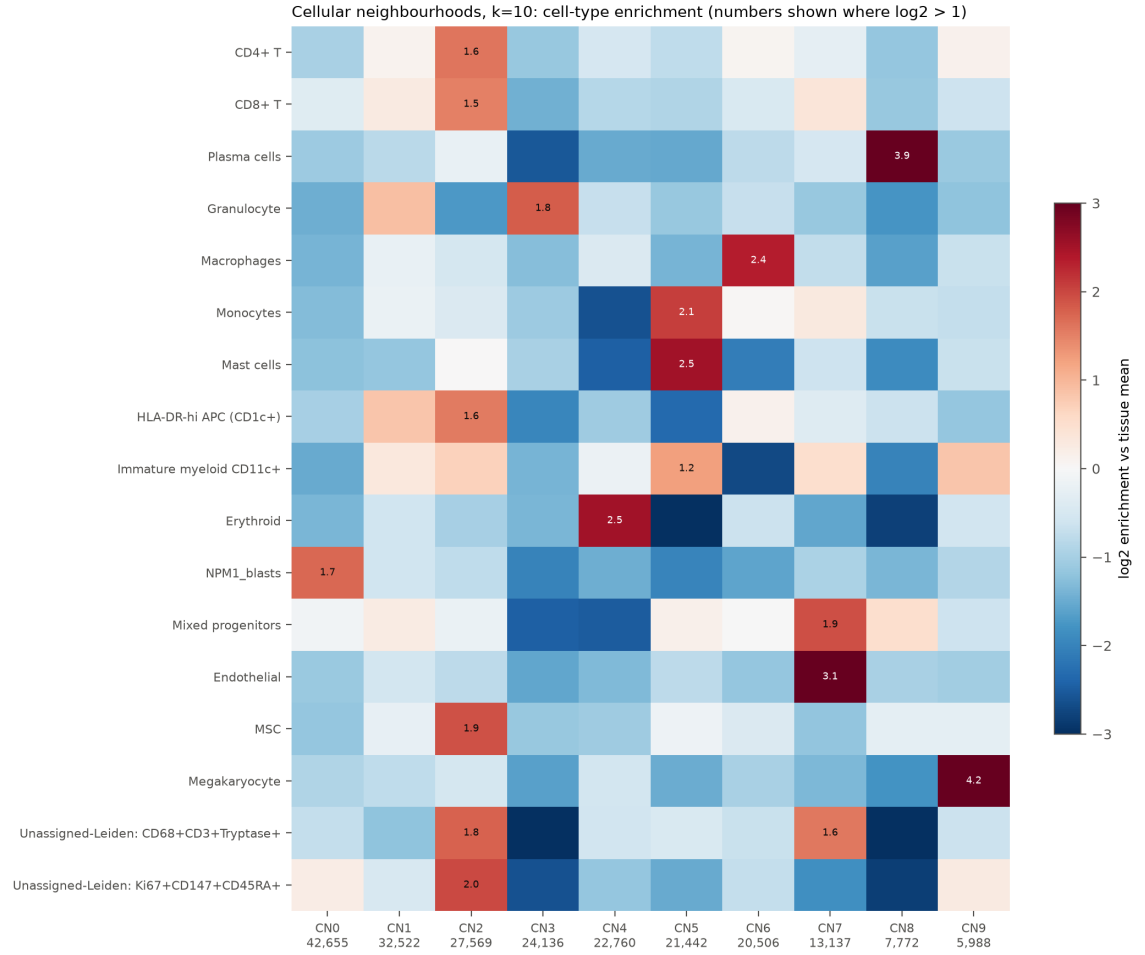
• **A vascular neighbourhood marks non-relapse.** The endothelial CN (CN7 at both k) covers 13.2% vs 1.5% of tissue (k = 10) and 17.2% vs 1.4% (k = 20), ROI BH ≤ 0.0003. This is the CN counterpart of the endothelial result in the proximity analysis.

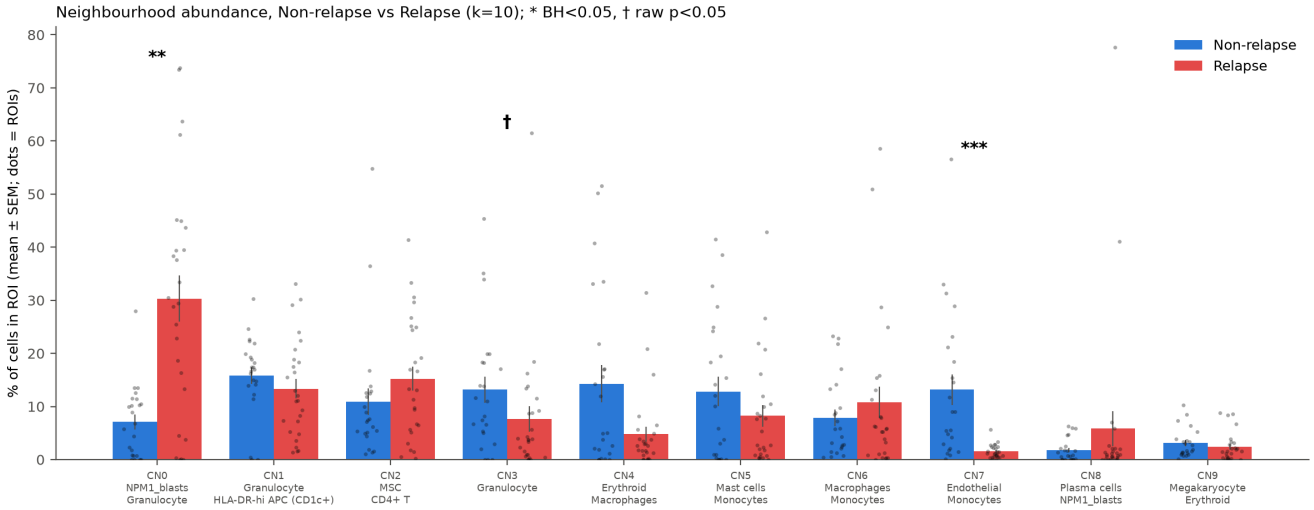
• **Healthy-haematopoiesis niches lean towards non-relapse but do not reach significance.** These are Granulocyte (CN3 at k = 10, CN0 at k = 20) and Erythroid (CN4 at k = 10, CN3 at k = 20), with raw p 0.03–0.06.

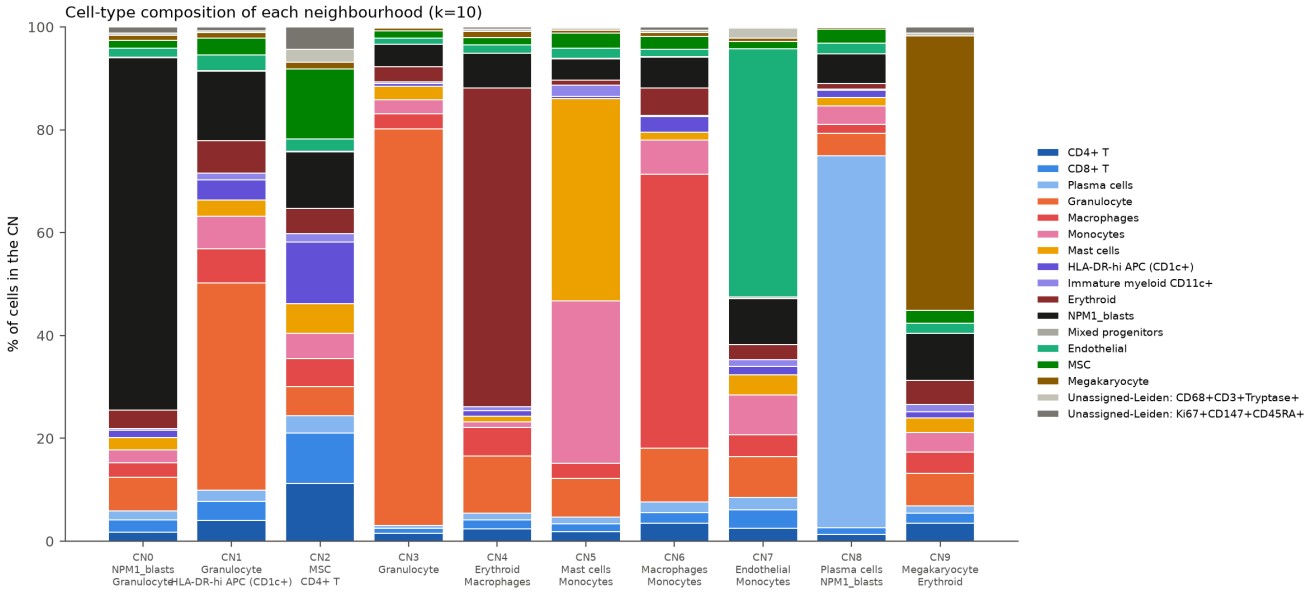
• **Patient-level confirmation.** With patients as the unit (6 vs 5), the blast + granulocyte CN (CN0 at k = 10, CN2 at k = 20) and the endothelial CN (CN7 at both k) remain significant (patient BH 0.022 each). The k = 20 blast + macrophage CN is borderline (patient p 0.052). Nothing else differs per patient.

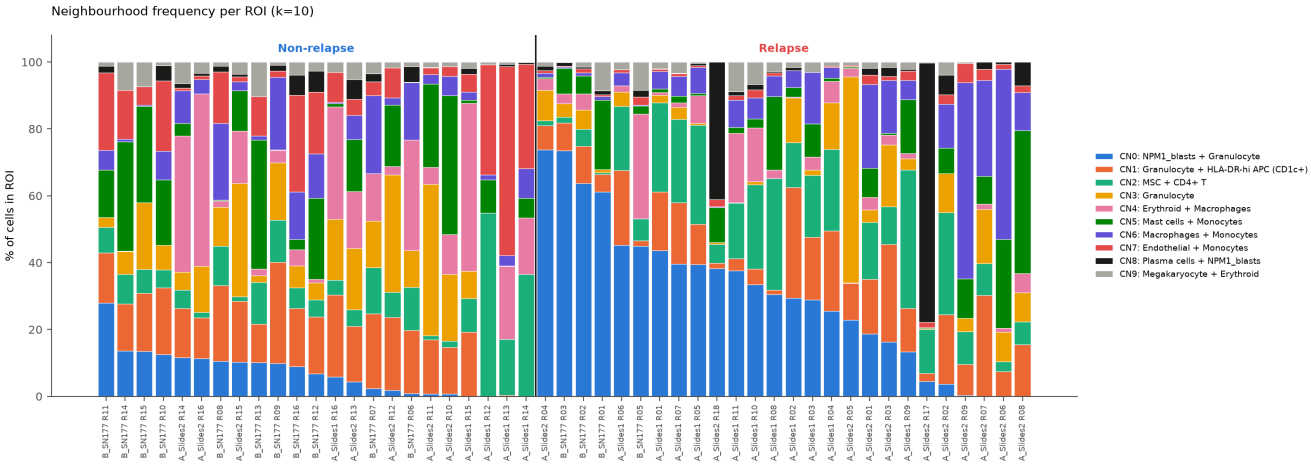
k = 10

CN	Name (top enriched types)	cells	NR %	R %	ROI BH p	patient p	patient BH
CN0	NPM1_blasts + Granulocyte	42,655	7.1	30.3	0.001	0.004	0.022
CN1	Granulocyte + HLA-DR-hi APC (CD1c+)	32,522	15.9	13.3	0.311	0.429	0.857
CN2	MSC + CD4+ T	27,569	10.9	15.2	0.232	0.662	0.931
CN3	Granulocyte	24,136	13.2	7.6	0.115	0.429	0.857
CN4	Erythroid + Macrophages	22,760	14.3	4.8	0.152	0.247	0.823
CN5	Mast cells + Monocytes	21,442	12.8	8.2	0.612	0.931	0.931
CN6	Macrophages + Monocytes	20,506	7.8	10.8	0.912	0.931	0.931
CN7	Endothelial + Monocytes	13,137	13.2	1.5	<0.001	0.004	0.022
CN8	Plasma cells + NPM1_blasts	7,772	1.8	5.8	0.856	0.792	0.931
CN9	Megakaryocyte + Erythroid	5,988	3.1	2.4	0.311	0.792	0.931


Cellular neighbourhoods, k=10: cell-type enrichment (numbers shown where log2 > 1)
CD4+ T
CD8+ T
Plasma cells
Granulocyte
Macrophages
Monocytes
Mast cells
HLA-DR-hi APC (CD1c+)
Immature myeloid CD11c+
Erythroid
NPM1_blasts
Mixed progenitors
Endothelial
MSC
Megakaryocyte
Unassigned-Leiden: CD68+CD3+Tryptase+
Unassigned-Leiden: Ki67+CD147+CD45RA+
1.6
1.5
3.9
1.8
2.4
2.1
2.5
1.6
1.2
2.5
1.7
1.9
3.1
1.9
4.2
1.8
1.6
2.0
CN0 42,655
CN1 32,522
CN2 27,569
CN3 24,136
CN4 22,760
CN5 21,442
CN6 20,506
CN7 13,137
CN8 7,772
CN9 5,988
log2 enrichment vs tissue mean
3
2
1
0
−1
−2
−3

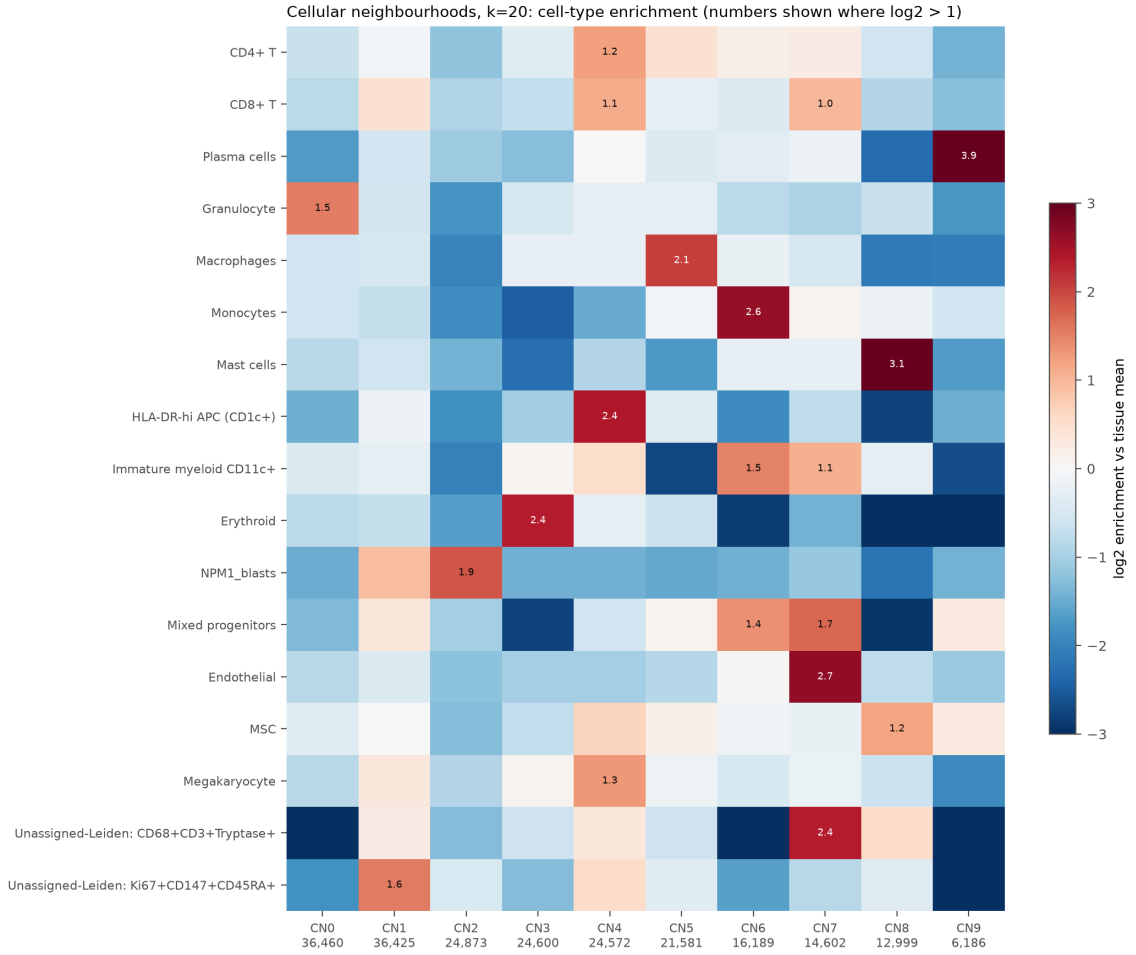

Neighbourhood abundance, Non-relapse vs Relapse (k=10); * BH<0.05, † raw p<0.05
% of cells in ROI (mean ± SEM; dots = ROIs)
Non-relapse
Relapse
**
†
***
0
10
20
30
40
50
60
70
80
CN0 NPM1_blasts Granulocyte
CN1 Granulocyte HLA-DR-hi APC (CD1c+)
CN2 MSC CD4+ T
CN3 Granulocyte
CN4 Erythroid Macrophages
CN5 Mast cells Monocytes
CN6 Macrophages Monocytes
CN7 Endothelial Monocytes
CN8 Plasma cells NPM1_blasts
CN9 Megakaryocyte Erythroid

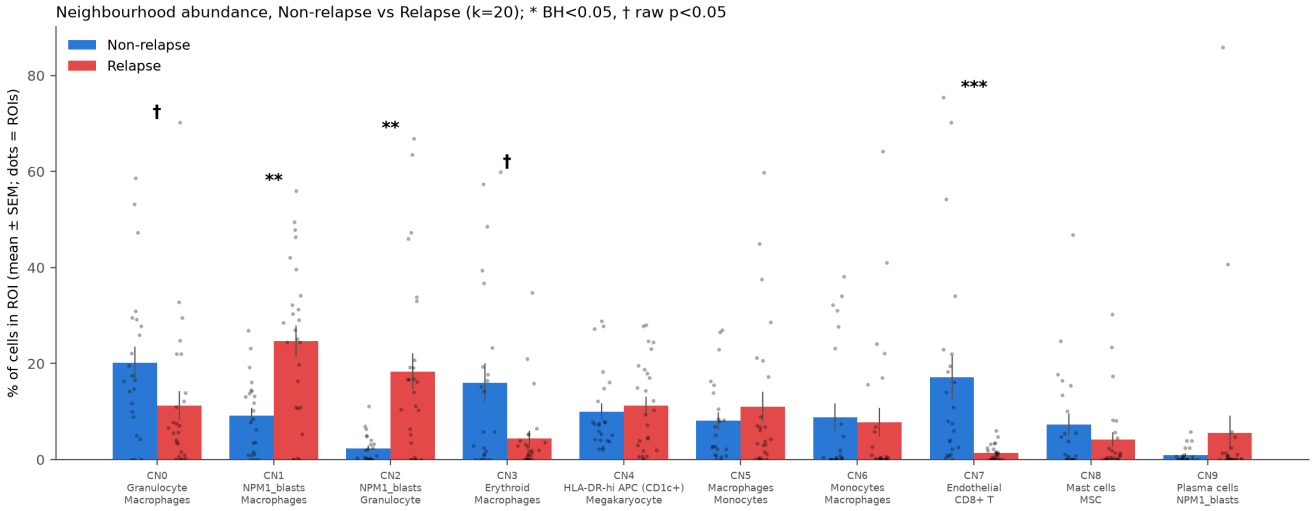

Cell-type composition of each neighbourhood (k=10)
% of cells in the CN
100
80
60
40
20
0
CN0
NPM1_blasts
Granulocyte
CN1
Granulocyte
HLA-DR-hi APC (CD1c+)
CN2
MSC
CD4+ T
CN3
Granulocyte
CN4
Erythroid
Macrophages
CN5
Mast cells
Monocytes
CN6
Macrophages
Monocytes
CN7
Endothelial
Monocytes
CN8
Plasma cells
NPM1_blasts
CN9
Megakaryocyte
Erythroid
CD4+ T
CD8+ T
Plasma cells
Granulocyte
Macrophages
Monocytes
Mast cells
HLA-DR-hi APC (CD1c+)
Immature myeloid CD11c+
Erythroid
NPM1_blasts
Mixed progenitors
Endothelial
MSC
Megakaryocyte
Unassigned-Leiden: CD68+CD3+Tryptase+
Unassigned-Leiden: Ki67+CD147+CD45RA+

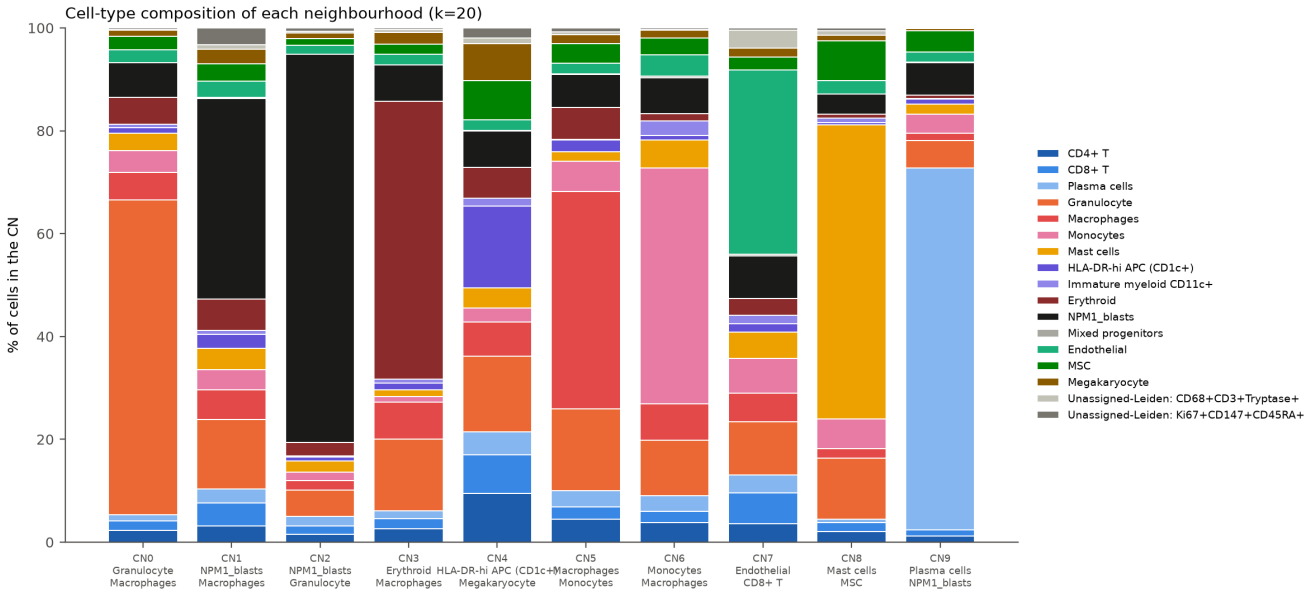

Neighbourhood frequency per ROI (k=10)
Non-relapse
Relapse
% of cells in ROI
100
80
60
40
20
0
CN0: NPM1_blasts + Granulocyte
CN1: Granulocyte + HLA-DR-hi APC (CD1c+)
CN2: MSC + CD4+ T
CN3: Granulocyte
CN4: Erythroid + Macrophages
CN5: Mast cells + Monocytes
CN6: Macrophages + Monocytes
CN7: Endothelial + Monocytes
CN8: Plasma cells + NPM1_blasts
CN9: Megakaryocyte + Erythroid

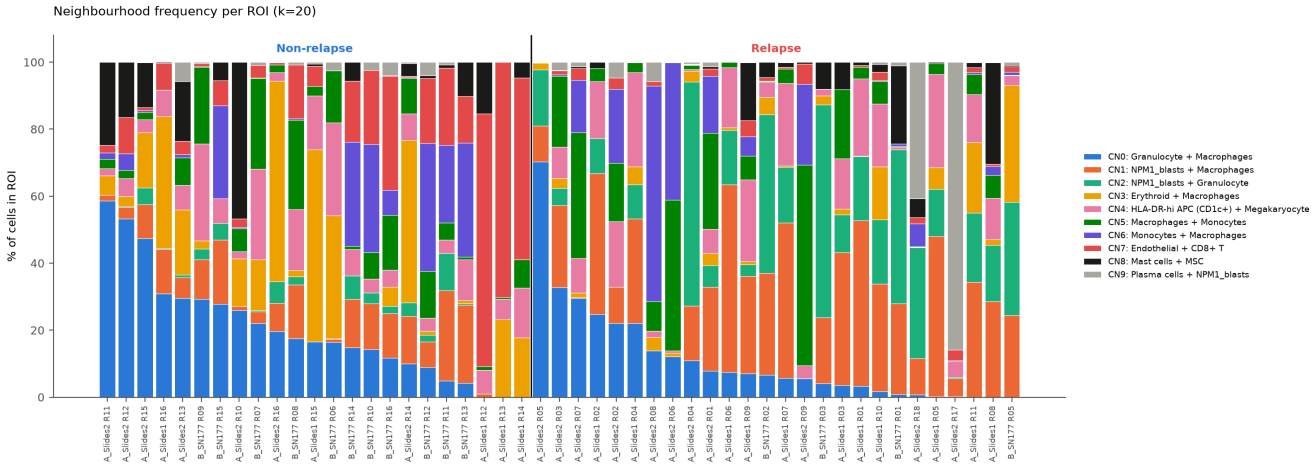
k = 20

CN	Name (top enriched types)	cells	NR %	R %	ROI BH p	patient p	patient BH
CN0	Granulocyte + Macrophages	36,460	20.2	11.3	0.067	0.247	0.494
CN1	NPM1_blasts + Macrophages	36,425	9.1	24.7	0.004	0.052	0.173
CN2	NPM1_blasts + Granulocyte	24,873	2.3	18.4	0.002	0.004	0.022
CN3	Erythroid + Macrophages	24,600	16.0	4.4	0.086	0.247	0.494
CN4	HLA-DR-hi APC (CD1c+) + Megakaryocyte	24,572	10.0	11.2	1.000	0.662	0.931
CN5	Macrophages + Monocytes	21,581	8.1	11.0	1.000	0.931	0.931
CN6	Monocytes + Macrophages	16,189	8.8	7.8	0.873	0.927	0.931
CN7	Endothelial + CD8+ T	14,602	17.2	1.4	<0.001	0.004	0.022
CN8	Mast cells + MSC	12,999	7.3	4.2	0.830	0.792	0.931
CN9	Plasma cells + NPM1_blasts	6,186	1.0	5.6	0.873	0.647	0.931

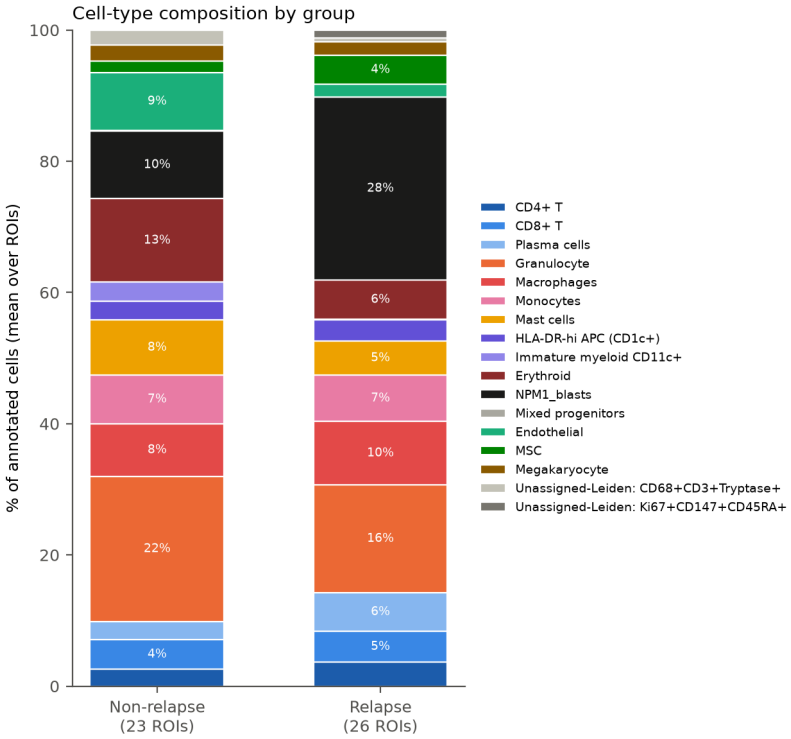



Neighbourhood abundance, Non-relapse vs Relapse (k=20); * BH<0.05, † raw p<0.05
Non-relapse
Relapse
% of cells in ROI (mean ± SEM; dots = ROIs)
0
20
40
60
80
†
**
**
†
***
CN0 Granulocyte Macrophages
CN1 NPM1_blasts Macrophages
CN2 NPM1_blasts Granulocyte
CN3 Erythroid Macrophages
CN4 HLA-DR-hi APC (CD1c+) Megakaryocyte
CN5 Macrophages Monocytes
CN6 Monocytes Macrophages
CN7 Endothelial CD8+ T
CN8 Mast cells MSC
CN9 Plasma cells NPM1_blasts


Cell-type composition of each neighbourhood (k=20)
% of cells in the CN
0
20
40
60
80
100
CN0 Granulocyte Macrophages
CN1 NPM1_blasts Macrophages
CN2 NPM1_blasts Granulocyte
CN3 Erythroid Macrophages
CN4 HLA-DR-hi APC (CD1c+) Megakaryocyte
CN5 Macrophages Monocytes
CN6 Monocytes Macrophages
CN7 Endothelial CD8+ T
CN8 Mast cells MSC
CN9 Plasma cells NPM1_blasts
CD4+ T
CD8+ T
Plasma cells
Granulocyte
Macrophages
Monocytes
Mast cells
HLA-DR-hi APC (CD1c+)
Immature myeloid CD11c+
Erythroid
NPM1_blasts
Mixed progenitors
Endothelial
MSC
Megakaryocyte
Unassigned-Leiden: CD68+CD3+Tryptase+
Unassigned-Leiden: Ki67+CD147+CD45RA+

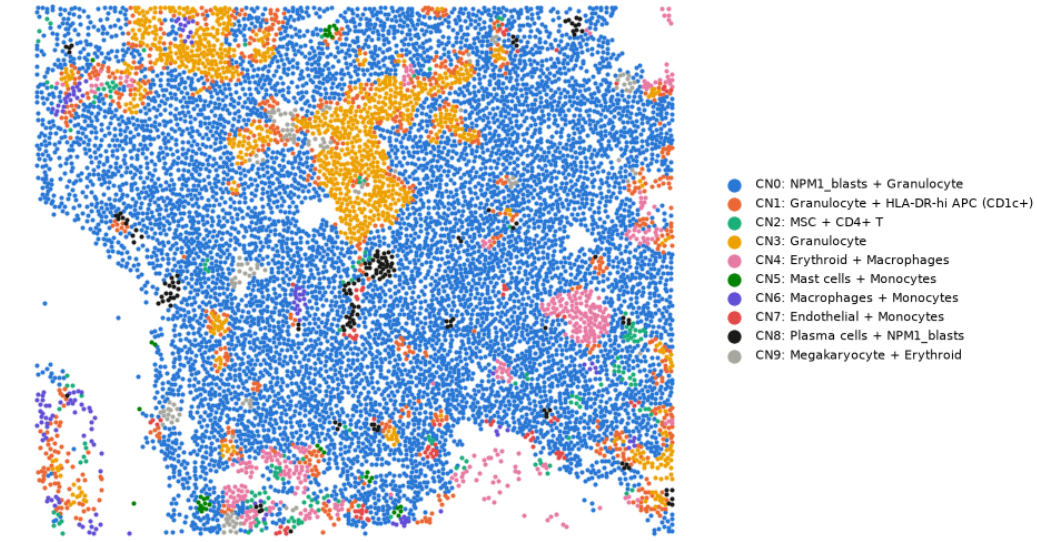


Overall cell-type composition by group

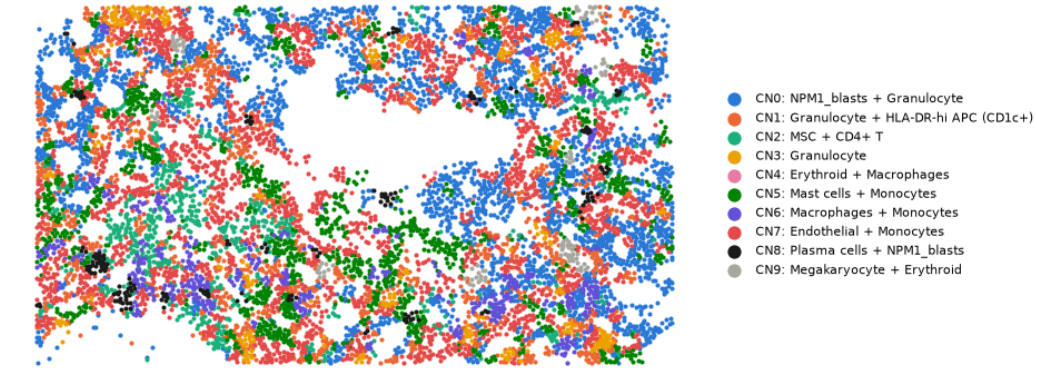


Example CN maps (k = 10)

Machine A/Slides2_ROI_04 — Relapse — CN k=10



Machine B/SN177_ROI_11 — Non-relapse — CN k=10



All 49 ROIs, for both k values, are in **k10/cn_maps/** and **k20/cn_maps/**.

Interpretation

The CN analysis gives the same tissue-level picture as the proximity analysis, using an independent method. In relapse marrow, a large share of the tissue is organised into blast-dominated neighbourhoods where blasts mix with granulocytes and macrophages. In non-relapse marrow, blasts are a minority and the tissue keeps a vascular (endothelial) neighbourhood and more normal myelo-erythroid niches. The endothelial-niche loss is the most consistent difference across the CN, proximity and composition analyses. A structured vascular niche may help keep residual disease under control. Its loss, or its replacement by blast-dominated tissue, accompanies relapse.

Clinically, the fraction of tissue in the blast CN and in the vascular CN are simple, per-biopsy spatial read-outs. They could complement MRD transcript levels for relapse prediction. This needs confirmation in an independent cohort, because the ROIs come from 11 patients and machine is partly confounded with group (relapse ROIs are mostly Machine A).

Viewing the CNs in MACSiQ View

MACSiQ_import/ holds one CSV per ROI (49 files). Each file has every cell of that ROI and these columns:

```
"Cell Id","Nuc X","Nuc Y Inv","CN_k10","CN_k20","CellType_int","Relapse"
```

- **Cell Id** is exactly as exported from MACSiQ View (for example "1@1"). This is the key MACSiQ View matches on. **Nuc X / Nuc Y Inv** are included so you can check that the right ROI is matched.
- **CN_k10 / CN_k20** are 0–9, or –1 for Unassigned/Artifact cells. **CellType_int** = final_label_int. **Relapse** = 1 for relapse, 0 for non-relapse. All values are numbers, so MACSiQ View treats them as features you can gate on. **_LEGEND_value_meanings.csv** lists what each number means.

Steps (menu names can differ between MACSiQ View versions; check the software manual for 'import features' if a label does not match):

1. Open the ROI's analysis in MACSiQ View, the same segmentation the original CSV was exported from.
2. Use the feature/data **import** option, choose that ROI's file (e.g. *Machine_A_Slides1_ROI_01_CN.csv*), and match on **Cell Id**.
3. The new columns appear as cell features. Make a histogram or scatter of **CN_k10**, gate one value at a time (e.g. 6.5–7.5 for CN7) and colour that gate on the image overlay.
4. Compare the overlay with **k10/cn_maps/<ROI>_CN_k10.png**. The patterns should match. If they look mirrored, the file was imported into the wrong ROI.

Files

build_CN_final.py / CN_final.ipynb is the full analysis (identical code). **k10/** and **k20/** hold cells_with_CN.csv, CN names, enrichment, composition, stats (ROI-level and patient-level), figs 1–4 and cn_maps/. **MACSiQ_import/** holds the per-ROI import CSVs and the legend.